Sync: a rotating-state SSM temporal block for EF estimation from echocardiography

Architectures of every model in the project, then side-by-side comparison with the baselines

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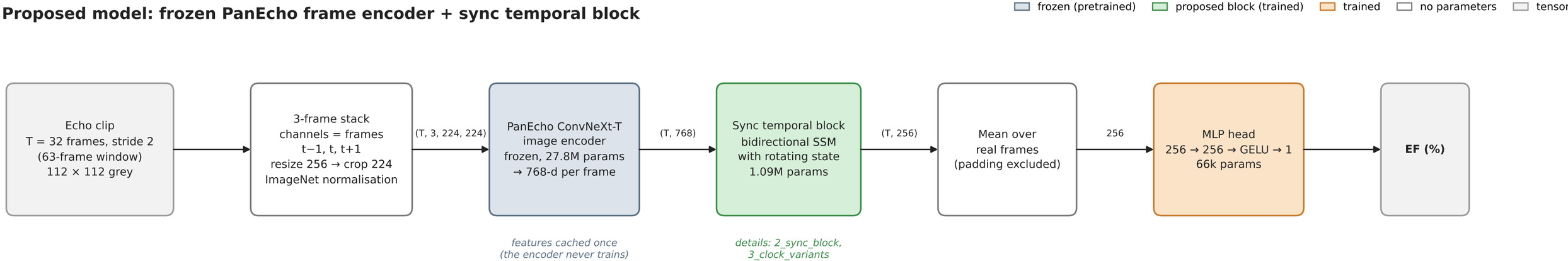
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Setup

- EchoNet-Dynamic, official split: 7,465 train / 1,288 validation / 1,277 test videos; EF label per video.
- Per-frame features from the frozen PanEcho ConvNeXt-T image encoder, each frame packed with its neighbours (t−1, t, t+1).
- Every temporal model: clips of 32 frames at stride 2, mean pool over real frames, the same MLP head, AdamW, 40 epochs, weight EMA.
- Baselines: BiGRU, Mamba, Transformer (UVT), no temporal model (same features and recipe); PanEcho's own temporal model (fine-tuned), ridge probe and zero-shot.
- All design decisions on the validation set; the test set evaluated once on the locked lineup. Test-set tables are included when the test evaluation has finished (included).

Data: EchoNet-Dynamic, Stanford AIMI (doi:10.71718/yqp5-y078), research use only. Encoder: PanEcho (Holste et al., JAMA 2025), weights CC BY-NC-SA 4.0.

Proposed model: frozen PanEcho frame encoder + sync temporal block

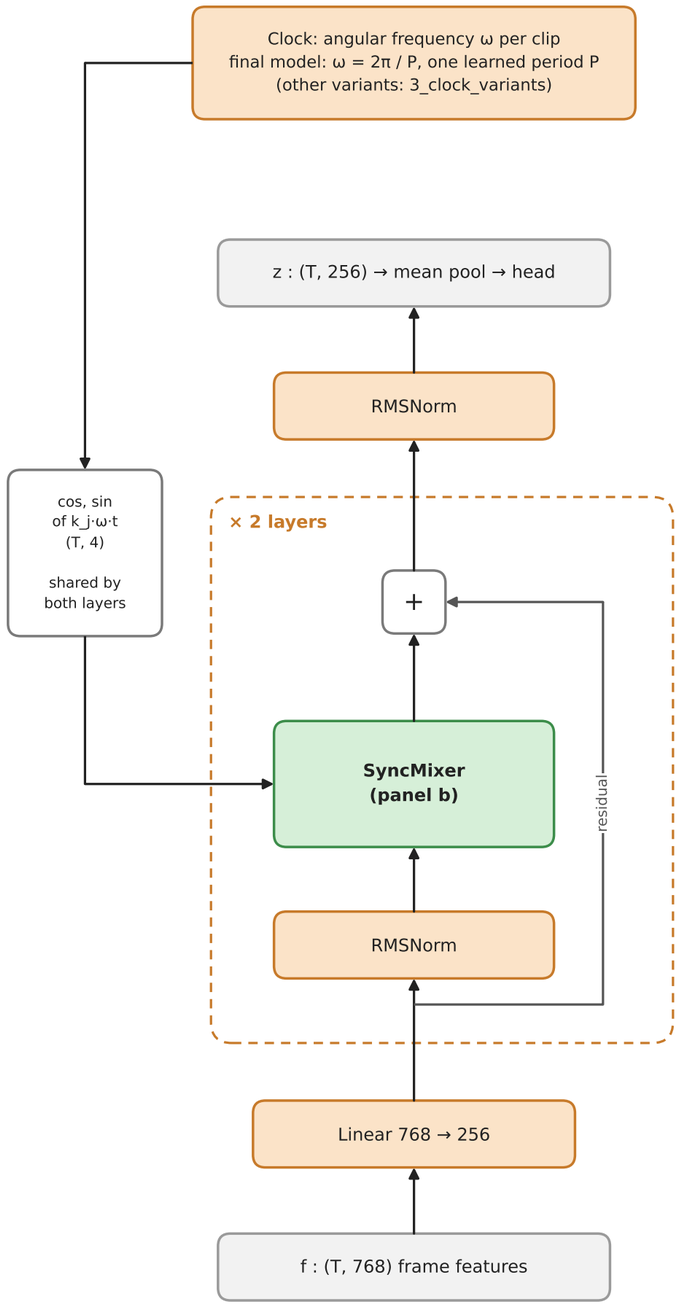


Training: MSE on standardised EF, AdamW lr 1e-3 (cosine, 5% warm-up), batch 16, 40 epochs; weight EMA 0.999 for validation, checkpoint selection and evaluation. Evaluation: mean EF over 4 evenly spaced clips per video.

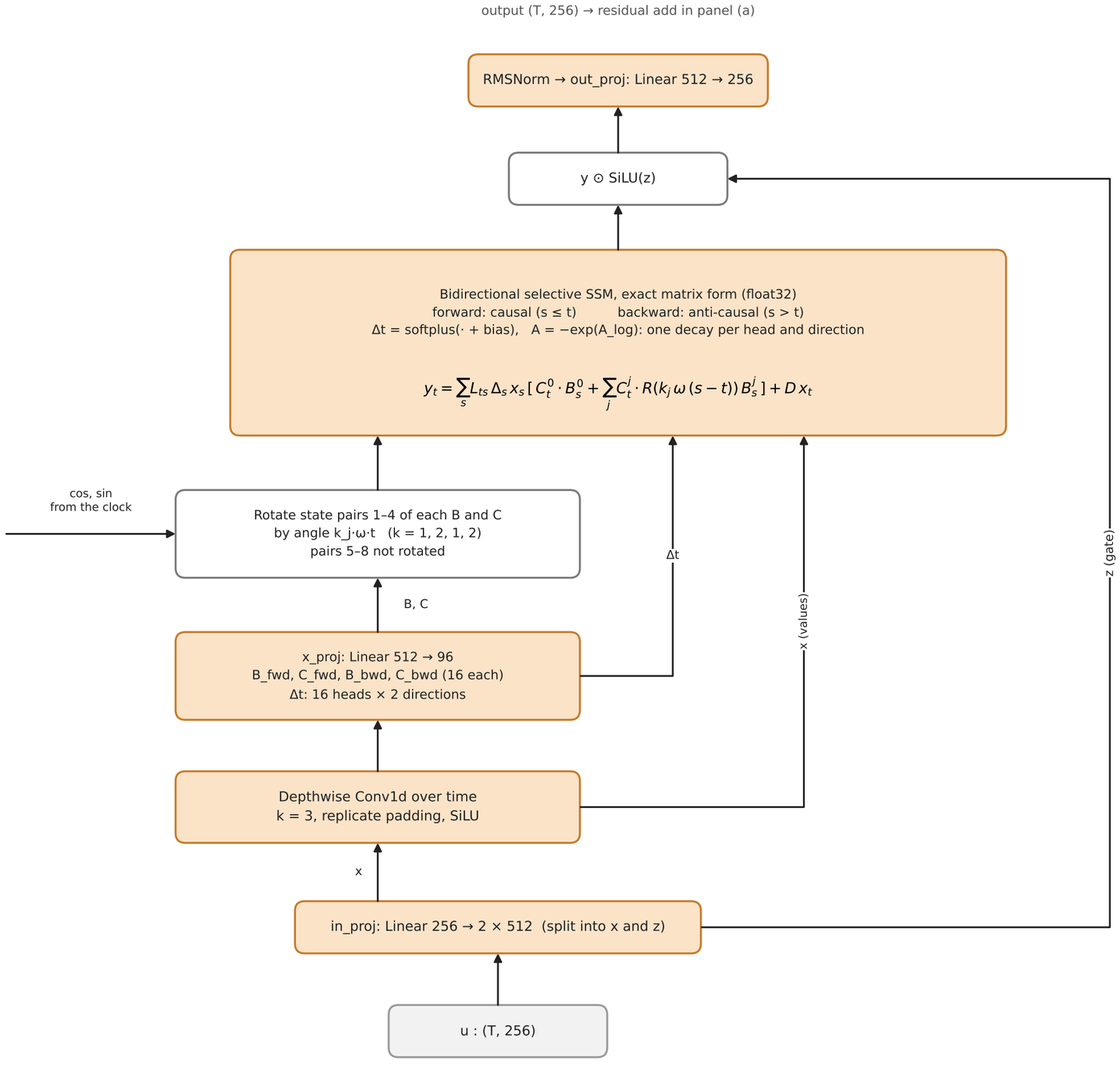
Sync temporal block (proposed)

trained · proposed block (trained) · no parameters · tensor

(a) SyncTemporal — 2 layers

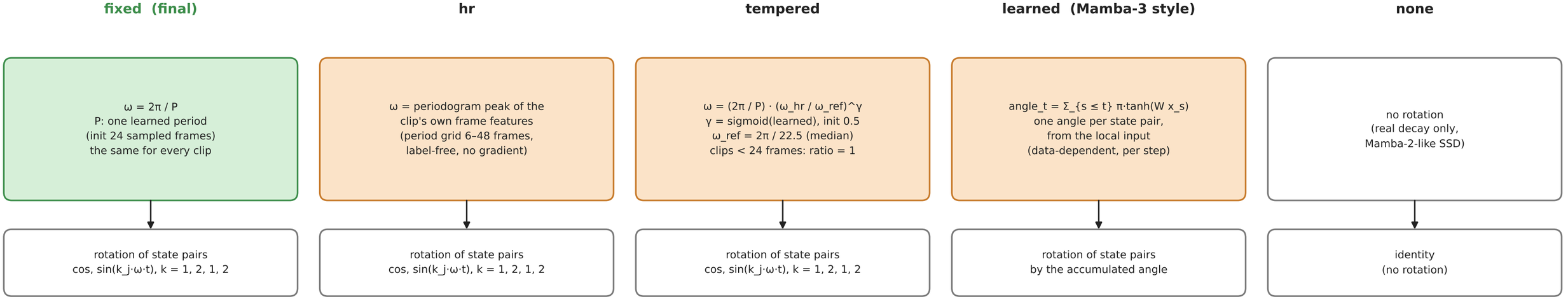


(b) SyncMixer — one layer: d_model 256, d_inner 512, 16 heads × 32 per direction, state N = 16



Where the rotation frequency of the sync block comes from

proposed block (trained) · trained · no parameters

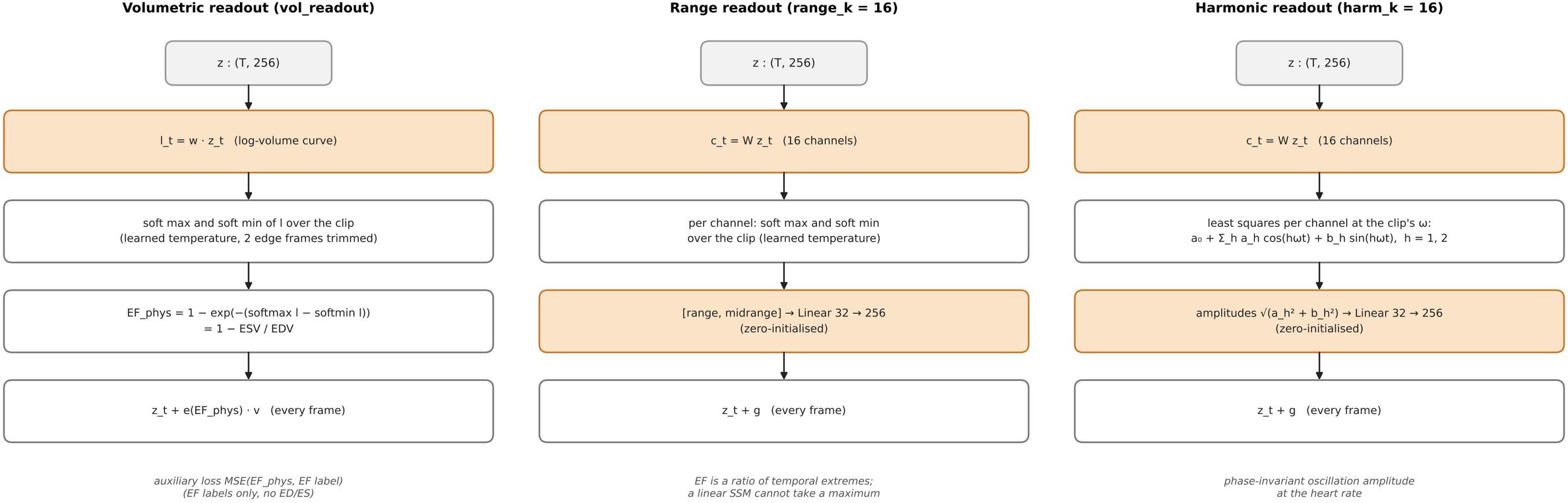


VAL MAE: 3 seeds, 4 clips per video, frozen PanEcho 3-frame-stack features, weight EMA. Everything else in the block is identical across the five variants.

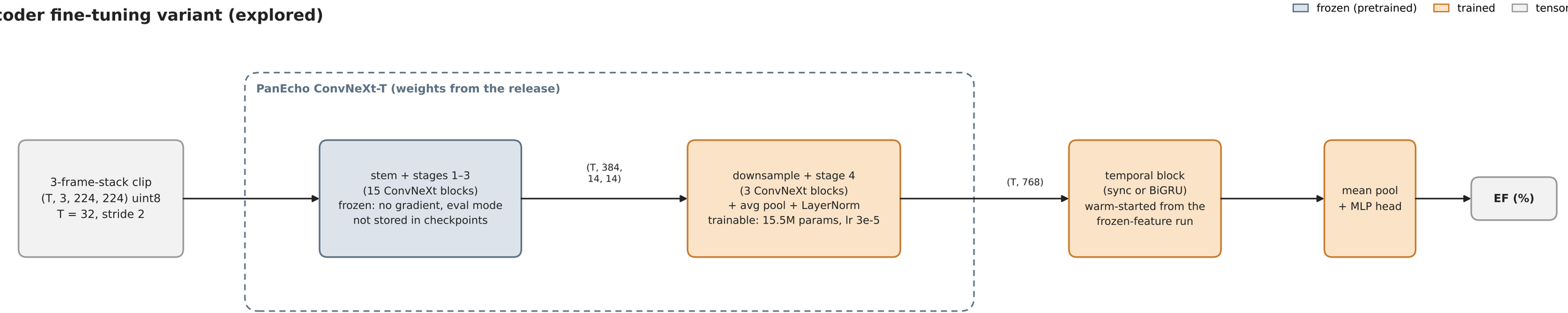
Explored readout heads on the sync block output

trained · no parameters · tensor

Each attaches to the block output z (after the final RMSNorm) and adds a vector to every frame, so the shared mean pool + head can use it. Explored; none was adopted (no VAL gain).

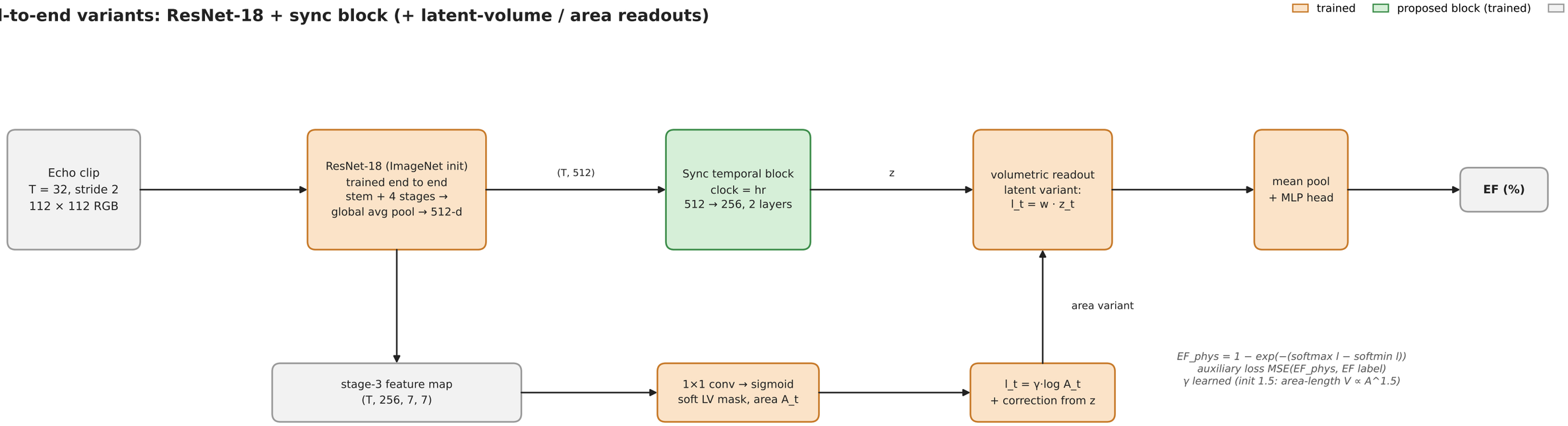


Encoder fine-tuning variant (explored)



Linear probe, then fine-tune (LP-FT): head lr 1e-4, encoder lr 3e-5, 5 epochs, weight EMA 0.999. Explored and not adopted: validation error rose every epoch (the encoder overfits 7,465 training videos).

End-to-end variants: ResNet-18 + sync block (+ latent-volume / area readouts)



First track (before the frozen-PanEcho track): same clip protocol, pooling and head as the proposed model; the area variant adds the mask branch.

Table 1. EF accuracy of all models — validation set

Every model uses the same frozen PanEcho ConvNeXt-T frame encoder.
proposed: sync temporal block variants | baselines on the same frozen features (+ weight EMA) | PanEcho baselines

	sync fixed (final)	sync hr clock	sync tempered	sync learned angles	sync no rotation	BiGRU	Mamba	Transformer (UVT)	no temporal model	PanEcho TF our clips	PanEcho TF 16 frames	PanEcho ridge probe	PanEcho zero-shot
trained parameters	1.15M	1.15M	1.15M	1.16M	1.15M	1.15M	1.14M	1.19M	0.26M	14.19M	14.19M	769	0
seeds	3	3	3	3	3	3	3	3	3	3	3	1	1
MAE (EF points) ↓	**3.963**	3.982	3.985	4.046	4.015	4.056	3.998	4.236	4.573	4.139	4.609	5.030	5.382
RMSE ↓	**5.253**	5.260	5.289	5.362	5.317	5.424	5.293	5.658	6.059	5.476	6.114	6.772	7.341
R^2 ↑	**0.818**	0.817	0.815	0.810	0.813	0.806	0.815	0.789	0.758	0.802	0.753	0.697	0.644
Pearson r ↑	**0.905**	0.904	0.903	0.900	0.902	0.900	0.903	0.890	0.871	0.896	0.869	0.838	0.807
CCC ↑	**0.899**	0.898	0.898	0.894	0.895	0.890	0.896	0.889	0.863	0.889	0.853	0.810	0.784
|bias| ↓	0.188	0.056	0.056	0.074	**0.019**	0.739	0.070	0.056	0.316	0.175	0.121	0.143	1.079
within 5 points (%) ↑	**71.1**	70.4	70.7	69.9	70.5	70.1	70.3	68.0	65.9	69.2	65.0	60.8	59.8
within 10 points (%) ↑	93.7	**93.8**	93.6	93.3	93.6	93.0	93.8	92.1	90.2	93.1	89.9	87.2	85.5
AUROC, EF < 40% ↑	**0.984**	0.984	0.984	0.984	0.984	0.983	0.983	0.981	0.978	0.982	0.976	0.969	0.959
AUROC, EF < 50% ↑	0.960	0.960	0.958	0.961	0.961	0.959	0.961	0.957	0.945	**0.961**	0.947	0.936	0.921
MAE difference vs BiGRU ↓	**-0.093**	-0.074	-0.071	-0.010	-0.041	reference	-0.057	+0.180	+0.517	+0.084	+0.553	+0.974	+1.326
95% CI of the difference	[-0.164, -0.020]	[-0.151, +0.001]	[-0.150, +0.007]	[-0.092, +0.073]	[-0.116, +0.034]	—	[-0.135, +0.019]	[+0.066, +0.290]	[+0.370, +0.663]	[-0.012, +0.176]	[+0.405, +0.705]	[+0.777, +1.174]	[+1.114, +1.544]
p vs BiGRU (paired bootstrap)	0.011	0.053	0.072	0.806	0.283	—	0.139	0.002	< 0.001	0.086	< 0.001	< 0.001	< 0.001

Validation set (1,288 videos), EF averaged over 4 evenly spaced clips per video, metrics averaged over seeds. Paired statistics over videos with seed-averaged absolute errors (10,000 bootstrap resamples).
Our clips: T = 32 frames at stride 2 (63-frame window); PanEcho's native clips: 16 consecutive frames. All models except the zero-shot one are trained on EchoNet-Dynamic TRAIN only; design choices made on validation.
Trained parameters: temporal model + head (the frozen PanEcho image encoder, 27.8M, is shared and excluded); ridge = 768 weights + bias. Bold = best in the row.

Table 1. EF accuracy of all models — test set

Every model uses the same frozen PanEcho ConvNeXt-T frame encoder.
proposed: sync temporal block variants baselines on the same frozen features (+ weight EMA) PanEcho baselines

	sync fixed (final)	sync hr clock	sync tempered	sync learned angles	sync no rotation	BiGRU	Mamba	Transformer (UVT)	no temporal model	PanEcho TF our clips	PanEcho TF 16 frames	PanEcho ridge probe	PanEcho zero-shot
trained parameters	1.15M	1.15M	1.15M	1.16M	1.15M	1.15M	1.14M	1.19M	0.26M	14.19M	14.19M	769	0
seeds	3	3	3	3	3	3	3	3	3	3	3	1	1
MAE (EF points) ↓	4.043	4.031	4.072	4.099	4.052	4.105	**4.020**	pending	pending	pending	pending	pending	pending
RMSE ↓	**5.302**	5.335	5.362	5.432	5.331	5.444	5.313	pending	pending	pending	pending	pending	pending
R^2 ↑	**0.812**	0.810	0.808	0.803	0.810	0.802	0.811	pending	pending	pending	pending	pending	pending
Pearson r ↑	**0.902**	0.900	0.899	0.896	0.901	0.900	0.901	pending	pending	pending	pending	pending	pending
CCC ↑	**0.894**	0.893	0.892	0.888	0.892	0.886	0.893	pending	pending	pending	pending	pending	pending
|bias| ↓	0.444	0.296	0.268	**0.163**	0.303	0.995	0.287	pending	pending	pending	pending	pending	pending
within 5 points (%) ↑	70.2	**71.0**	70.8	70.5	70.8	69.4	70.7	pending	pending	pending	pending	pending	pending
within 10 points (%) ↑	93.2	**93.4**	93.3	92.9	92.9	93.0	93.4	pending	pending	pending	pending	pending	pending
AUROC, EF < 40% ↑	**0.983**	0.981	0.980	0.980	0.981	0.981	0.982	pending	pending	pending	pending	pending	pending
AUROC, EF < 50% ↑	0.966	0.965	0.964	0.965	0.967	0.967	**0.967**	pending	pending	pending	pending	pending	pending
MAE difference vs BiGRU ↓	-0.062	-0.074	-0.033	-0.006	-0.053	reference	**-0.085**	pending	pending	pending	pending	pending	pending
95% CI of the difference	[-0.132, +0.011]	[-0.150, +0.004]	[-0.111, +0.047]	[-0.087, +0.075]	[-0.123, +0.019]	—	[-0.159, -0.007]	pending	pending	pending	pending	pending	pending
p vs BiGRU (paired bootstrap)	0.093	0.061	0.411	0.895	0.157	—	0.032	pending	pending	pending	pending	pending	pending

Test set (1,277 videos), EF averaged over 4 evenly spaced clips per video, metrics averaged over seeds. Paired statistics over videos with seed-averaged absolute errors (10,000 bootstrap resamples).
Our clips: T = 32 frames at stride 2 (63-frame window); PanEcho's native clips: 16 consecutive frames. All models except the zero-shot one are trained on EchoNet-Dynamic TRAIN only; design choices made on validation.
Trained parameters: temporal model + head (the frozen PanEcho image encoder, 27.8M, is shared and excluded); ridge = 768 weights + bias. Bold = best in the row.

Table 2. Temporal behaviour and cost — validation set

Proposed variants and baselines on the same frozen PanEcho 3-frame-stack features.

proposed: sync temporal block variants | baselines on the same frozen features (+ weight EMA) | PanEcho baselines

	sync fixed (final)	sync hr clock	sync tempered	sync learned angles	sync no rotation	BiGRU	Mamba	Transformer (UVT)	no temporal model	PanEcho TF our clips
MAE at T = 8 ↓	5.998	7.843	6.329	6.471	6.358	**5.861**	8.520	11.241	6.626	—
MAE at T = 16 ↓	**4.288**	4.562	4.323	4.407	4.408	4.504	4.615	5.036	4.797	—
MAE at T = 32 (trained) ↓	**3.963**	3.982	3.985	4.046	4.015	4.056	3.998	4.236	4.573	—
MAE at T = 64 ↓	**3.989**	4.003	3.994	4.037	4.011	4.024	4.009	4.339	4.495	—
MAE at T = 128 ↓	4.112	4.088	4.082	4.071	4.081	**4.064**	4.091	4.428	4.485	—
MAE, heart rate < 60 bpm (n = 321) ↓	3.648	**3.633**	3.676	3.729	3.709	3.636	3.708	3.915	4.290	—
MAE, heart rate 60–100 bpm (n = 884) ↓	**4.013**	4.060	4.053	4.099	4.048	4.130	4.040	4.277	4.558	—
MAE, heart rate > 100 bpm (n = 83) ↓	4.656	4.491	**4.458**	4.702	4.842	4.897	4.676	5.032	5.819	—
EF SD across a video's 4 clips ↓	1.476	1.449	1.513	1.543	1.443	1.345	1.407	1.537	**1.330**	—
feature smoothness, cos(z_t, z_t+1) (descriptive)	0.890	0.888	0.896	0.838	0.846	0.926	0.812	0.968	0.898	—
periodicity, autocorrelation peak (descriptive)	0.185	0.230	0.199	0.224	0.219	0.144	0.192	0.229	0.237	—
ED frame probe error (frames) ↓	2.125	2.155	2.119	2.293	2.228	2.093	2.239	**1.968**	2.649	—
ES frame probe error (frames) ↓	2.128	2.052	2.103	2.278	2.138	2.315	2.129	**1.793**	2.391	—
MAE with frames shuffled (higher = uses order)	14.835	16.048	15.465	18.894	18.419	15.468	15.455	4.245	4.573	—
forward GFLOP per clip ↓	0.072	0.087	—	—	—	0.069	0.068	0.069	**0.013**	0.919
latency, ms per clip (batch 16) ↓	0.082	0.131	—	—	—	0.040	1.101	0.026	**0.007**	0.334
training step, ms (batch 16) ↓	10.6	11.3	—	—	—	3.6	56.6	5.6	**1.3**	16.4

Validation set, mean of 3 seeds, 4 clips per video. Models trained once at T = 32 (stride 2) and evaluated at every T. Heart rate: label-free whole-video estimate from the frame features (scripts/hr_reference.py).
Probes: ridge regression from the frozen block outputs to the ED/ES frame (fitted on TRAIN, lambda on validation); error in sampled frames. Smoothness and periodicity are descriptive, not accuracy.
Cost: temporal model + head only (the shared frozen encoder is excluded), RTX 3050 laptop GPU, bf16, CUDA graphs; '—' = not measured or not applicable (PanEcho TF has no per-frame evaluation outputs).

Table 2. Temporal behaviour and cost — test set

Proposed variants and baselines on the same frozen PanEcho 3-frame-stack features.

proposed: sync temporal block variants | baselines on the same frozen features (+ weight EMA) | PanEcho baselines

	sync fixed (final)	sync hr clock	sync tempered	sync learned angles	sync no rotation	BiGRU	Mamba	Transformer (UVT)	no temporal model	PanEcho TF our clips
MAE at T = 8 ↓	5.917	7.709	6.261	6.456	6.262	**5.795**	8.470	pending	pending	—
MAE at T = 16 ↓	**4.381**	4.513	4.396	4.528	4.436	4.675	4.595	pending	pending	—
MAE at T = 32 (trained) ↓	4.043	4.031	4.072	4.099	4.052	4.105	**4.020**	pending	pending	—
MAE at T = 64 ↓	4.064	4.081	4.074	4.127	4.088	**4.043**	4.085	pending	pending	—
MAE at T = 128 ↓	4.140	4.181	4.134	4.163	4.138	**4.049**	4.198	pending	pending	—
MAE, heart rate < 60 bpm (n = 296) ↓	3.926	3.958	4.012	3.945	3.944	4.030	**3.874**	pending	pending	—
MAE, heart rate 60–100 bpm (n = 904) ↓	4.043	**4.018**	4.054	4.082	4.030	4.065	4.018	pending	pending	—
MAE, heart rate > 100 bpm (n = 77) ↓	4.505	**4.463**	4.509	4.881	4.729	4.865	4.599	pending	pending	—
EF SD across a video's 4 clips ↓	1.466	1.453	1.498	1.533	1.423	**1.341**	1.390	pending	pending	—
feature smoothness, cos(z_t, z_t+1) (descriptive)	0.890	0.887	0.896	0.839	0.845	0.925	0.811	pending	pending	—
periodicity, autocorrelation peak (descriptive)	0.185	0.231	0.200	0.225	0.220	0.144	0.193	pending	pending	—
ED frame probe error (frames) ↓	**2.119**	2.206	2.170	2.338	2.283	2.211	2.377	pending	pending	—
ES frame probe error (frames) ↓	2.269	**2.093**	2.283	2.351	2.243	2.414	2.255	pending	pending	—
MAE with frames shuffled (higher = uses order)	14.699	15.919	15.301	18.772	18.173	15.539	15.360	pending	pending	—
forward GFLOP per clip ↓	0.072	0.087	—	—	—	0.069	0.068	0.069	**0.013**	0.919
latency, ms per clip (batch 16) ↓	0.082	0.131	—	—	—	0.040	1.101	0.026	**0.007**	0.334
training step, ms (batch 16) ↓	10.6	11.3	—	—	—	3.6	56.6	5.6	**1.3**	16.4

Test set, mean of 3 seeds, 4 clips per video. Models trained once at T = 32 (stride 2) and evaluated at every T. Heart rate: label-free whole-video estimate from the frame features (scripts/hr_reference.py).
Probes: ridge regression from the frozen block outputs to the ED/ES frame (fitted on TRAIN, lambda on validation); error in sampled frames. Smoothness and periodicity are descriptive, not accuracy.
Cost: temporal model + head only (the shared frozen encoder is excluded), RTX 3050 laptop GPU, bf16, CUDA graphs; '—' = not measured or not applicable (PanEcho TF has no per-frame evaluation outputs).

Table 3. Explored variants vs BiGRU and Mamba under matched protocols — validation set

MAE in EF points (lower is better). Each variant is compared only with baselines trained and evaluated exactly as it was; values are not comparable across protocols.

family	variant	protocol (same for the variant and its baselines)	variant	BiGRU	Mamba	Δ vs BiGRU
Readout heads	sync, no readout	single-frame PanEcho features; seed 0; best single-clip VAL MAE (training log)	4.676	4.547	—	+0.129
	sync + volumetric readout	single-frame PanEcho features; seed 0; best single-clip VAL MAE (training log)	4.624	4.547	—	+0.077
	sync + harmonic + range readouts	single-frame PanEcho features; seed 0; best single-clip VAL MAE (training log)	4.704	4.547	—	+0.157
	BiGRU + harmonic + range readouts	single-frame PanEcho features; seed 0; best single-clip VAL MAE (training log)	4.695	4.547	—	+0.148
	sync, no readout	3-frame-stack features, no EMA; 3 seeds; 4-clip VAL MAE	4.063	4.101	4.098	**-0.038**
	sync + volumetric readout	3-frame-stack features, no EMA; 3 seeds; 4-clip VAL MAE	4.194	4.101	4.098	+0.093
Encoder fine-tuning	BiGRU, encoder stage 4 fine-tuned	3-frame-stack clips; seed 0; 4-clip VAL MAE (baselines frozen, EMA)	4.013	4.036	3.998	**-0.023**
End to end	sync (hr clock) + latent volume	ResNet-18 trained end to end; seed 0; best single-clip VAL MAE (training log)	4.573	4.474	4.595	+0.099
	sync (hr clock) + area readout	ResNet-18 trained end to end; seed 0; best single-clip VAL MAE (training log)	4.435	4.474	4.595	**-0.039**

Readout heads: 'sync + range' alone did not complete (1 of 40 epochs) and is omitted. Δ in bold = better than BiGRU. Single-seed rows carry seed noise of about ±0.03-0.05 MAE.
Encoder fine-tuning: best epoch is the first; validation error rose in every later epoch (5_encoder_finetuning). End to end: the first project track, before the frozen-PanEcho features.
None of these variants was adopted; the proposed model is the sync block with a fixed learned frequency on frozen 3-frame-stack features (Tables 1-2).